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Experiment M3-bis2
Coupled Oscillations

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1 Introduction

The study of coupled oscillations is fundamental in physics, as it provides a framework for understanding how energy is transferred between different degrees of freedom in complex mechanical systems. In this laboratory session, we investigate the dynamics of a rigid bar suspended by two identical springs, a system that exhibits inertial and elastic coupling [1].

The motion of such a system can be described as a linear combination of its normal modes of oscillation. These modes are specific patterns of motion where all parts of the system oscillate with the same characteristic frequency and a fixed phase relationship. For this experimental setup, two primary modes are expected:

- **Symmetric Mode (ω_s):** Where the bar translates vertically without rotation, meaning both ends move in phase.
- **Antisymmetric Mode (ω_a):** Where the bar rotates about its center of mass, and the ends move in opposite directions (out of phase).

We can define two objetives for this experiment. First, to identify these natural frequencies through direct observation and sinusoidal fitting. Second, to analyze complex oscillations resulting from the superposition of these modes using advanced signal processing techniques, such as the Fast Fourier Transform (FFT). Furthermore, we aim to demonstrate the decoupling of the equations of motion by transforming the recorded force data into collective coordinates (z_{CM} and z_{12}), which allows for the isolation of each fundamental mode regardless of the initial perturbation.

By comparing experimental results with theoretical predictions, we will also evaluate the impact of real-world factors, such as the exact positioning of the springs, on the system's characteristic frequencies.

2 Results and discussion

2.1 Determination of the period and frequency in pure harmonic oscillations

The objective of this section is to calculate the system's oscillation period using three complementary methodologies, therefore evaluating the consistency of the experimental results.

Firstly, a descriptive statistical analysis was performed on the direct readings provided by the sensor. To ensure the reliability of this calculation, only temporal intervals that showed the greatest stability were selected, discarding initial transients and residual noise. The average period value was then derived from this stable subset of data.

Secondly, a fitting of the raw force-versus-time data to a sinusoidal model was used, both for the left and right springs. This procedure allows us to determine the angular frequency (ω) and, by extension, the period, while minimizing the impact of random fluctuations and experimental uncertainties.

Finally, we implemented an algorithm based on the Fast Fourier Transform (FFT). By locating the maximum in the generated spectrum, we determined the frequency (f), from which the period (T) and angular frequency (ω) were calculated. The following section details and compares these parameters across the different oscillation modes.

Looking at the results, there is a remarkable consistency across all oscillation modes, with a relative error of $\varepsilon < 1$. The values for period, frequency, and angular frequency remain nearly identical regardless of whether we used experimental measurements, sinusoidal fitting, or FFT analysis. This high level of agreement between the three independent methods confirms that the results are highly reliable.

2.1.1 Oscillation with a fixed end

This section provides the average parameters obtained for the three experiences realized. The complete data calculated can be found in Table 7, 8 and 9.

	$\overline{T_o}$ (s)	ω_o (rad/s)	f_o (Hz)
Experimental	$0,4314 \pm 0,0065$	$14,57 \pm 0,22$	$2,318 \pm 0,035$
Sinusoidal	$0,42963 \pm 0,0001$	$14,6245 \pm 0,0036$	$2,3276 \pm 0,0005$
FFT	$0,4285 \pm 0,0033$	$14,6632 \pm 0,0020$	$2,33372 \pm 0,00032$

Table 1: Comparison of period $\overline{T_o}$, angular frequency ω_o and f_o for the first oscillation mode, determined via experimental measurement, sinusoidal fitting, and Fast Fourier Transform (FFT).

2.1.2 Symmetrical oscillation

Below, we present the average results from the three experiences for the symmetrical oscillation mode. For the complete breakdown of the data, please refer to Tables 10, 11, and 12.

	$\overline{T_s}$ (s)	ω_s (rad/s)	f_s (Hz)
Experimental	$0,4887 \pm 0,0012$	$12,856 \pm 0,031$	$2,0461 \pm 0,0049$
Sinusoidal	$0,491483 \pm 0,000076$	$12,7841 \pm 0,002$	$2,03466 \pm 0,00032$
FFT	$0,4941 \pm 0,0033$	$12,7171 \pm 0,0014$	$2,02399 \pm 0,00023$

Table 2: Comparison of period $\overline{T_s}$, angular frequency ω_s and f_s for the symmetric oscillation mode.

2.1.3 Antisymmetric oscillation

In this case, we summarize the average parameters from the three experiments of the antisymmetric oscillation mode. Tables 13, 14, and 15 present extensive data on this subject.

	$\overline{T_a}$ (s)	ω_a (rad/s)	f_a (Hz)
Experimental	$0,3322 \pm 0,0025$	$18,92 \pm 0,13$	$3,011 \pm 0,021$
Sinusoidal	$0,32903 \pm 0,00022$	$19,096 \pm 0,013$	$3,0390 \pm 0,0020$
FFT	$0,3295 \pm 0,0022$	$19,0704 \pm 0,0014$	$3,03514 \pm 0,00023$

Table 3: Comparison of period $\overline{T_a}$, angular frequency ω_a and f_a for the antisymmetric oscillation mode.

2.2 Angular frequencies ω_o, ω_s and ω_a

The characteristic parameters of the observed harmonic motion, including the angular frequencies ω_o, ω_s, and ω_a along with their respective uncertainties, are compiled in the table below. This allows for a direct comparative analysis between the values derived from manual experimental sampling and those obtained through sinusoidal regression.

	ω_o (rad/s)	ω_s (rad/s)	ω_a (rad/s)
Experimental method	$14,57 \pm 0,22$	$12,856 \pm 0,031$	$18,92 \pm 0,13$
Sinusoidal regression	$14,6245 \pm 0,0036$	$12,7841 \pm 0,0020$	$19,096 \pm 0,013$
FFT	$14,6632 \pm 0,0020$	$12,7171 \pm 0,0014$	$19,0704 \pm 0,0014$
Average value	$14,619 \pm 0,073$	$12,786 \pm 0,010$	$19,030 \pm 0,044$

Table 4: Angular frequencies ω_o, ω_s and ω_a with their corresponding uncertainties, obtained using the experimental method, the sinusoidal fitting and the Fast Fourier Transform.

As can be seen in Table 4, the angular frequencies clearly follow a hierarchy, with $\omega_s < \omega_o < \omega_a$. From a theoretical perspective, by solving the characteristic equations of the system, the relationship between the normal modes of oscillation is established as follows:

$$\omega_a = \sqrt{3} \cdot \sqrt{\frac{2k}{m}} = \sqrt{3} \cdot \omega_s \Rightarrow \frac{\omega_a}{\omega_s} \approx 1,732. \tag{1}$$

Nevertheless, the experimental ratio between ω_a and ω_s yields a value of approximately 1.5, revealing a deviation from the ideal model. This discrepancy suggests that the theoretical approach was not precise. Specifically, the model assumes that the elastic forces are applied exactly at the distal ends of the bar. However, in the experimental setup the springs were positioned at a small, but significant, distance from the boundaries. This internal displacement of the coupling points effectively reduces the restoring torque during the antisymmetric mode. As a result, ω_a becomes lower than predicted, bringing it closer to the frequency of the symmetric mode.

2.3 Superposition of Modes and Analysis of Natural Frequencies

This section investigates the system's behavior when subjected to complex motion through the superposition of its normal modes. Following the experimental procedure, one end of the bar was displaced while the other was held at rest to simultaneously excite both the symmetric and antisymmetric modes [1].

2.3.1 Experimental Results

To determine the natural frequencies ω_{min} and ω_{max}, a Fast Fourier Transform (FFT) was applied to the force signals recorded by the PASCO sensors at both ends of the bar. The results obtained from the various trials are presented in the following table, compared with the reference values obtained in Section 2 ($\omega_s = 12.856$ rad/s and $\omega_a = 18.92$ rad/s):

Trial, Sensor (Force)	**FFT Res.** (rad/s)	ω_{min} (rad/s)	**Error** vs ω_s	ω_{max} (rad/s)	**Error** vs ω_a
1, RIGHT (R)	0.1853	12.7873	0.53%	19.0883	0.89%
1, RIGHT (L)	0.1852	12.7812	0.58%	19.0792	0.84%
*1, LEFT (R)**	0.1382	12.7173	1.08%	12.8556	32.05%
*1, LEFT (L)**	0.1382	12.7128	1.11%	12.8510	32.08%
2, RIGHT (R)	0.2095	12.7758	0.62%	19.2684	1.84%
2, RIGHT (L)	0.2095	12.7758	0.62%	19.0590	0.73%
2, LEFT (R)	0.2095	12.7758	0.62%	19.0590	0.73%
2, LEFT (L)	0.2095	12.7758	0.62%	19.0590	0.73%

Table 5: Natural frequencies identified via FFT. (R) denotes Right and (L) denotes Left.

2.3.2 Analysis and Discussion

According to the results, the identified minimum frequency (ω_{min}) shows excellent agreement with the symmetric mode frequency (ω_s), while the maximum frequency (ω_{max}) corresponds to the antisymmetric mode (ω_a). In the trials initiated from the right end (RIGHT), the relative errors remain below 2%, validating the theoretical principle that any complex motion can be described as a linear combination of the system's normal modes. This superposition of frequencies is visually confirmed in the spectral analysis shown in the graphs generated (see drive [2]), where both characteristic peaks are clearly distinguishable.

On the other hand, an atypical error exceeding 32% is observed in the determination of ω_{max} for the trial "**1, LEFT**" datasets. We can identify the origin of this error as the fact that the the antisymmetric mode was not sufficiently excited, preventing the FFT algorithm from identifying ω_{max} correctly in that specific sensor signal. Consequently, the spectral resolution (0.1382 rad/s) and the peak-detection algorithm failed to distinguish a second clear frequency peak, resulting in an erroneous reading close to the symmetric frequency itself.

As a conclusion, we can say that the simultaneous detection of both frequency peaks in a single sensor confirms the existence of mechanical inertial coupling in the rigid body. As shown, the system transfers energy between its degrees of freedom such that, regardless of where the initial perturbation occurs, the bar oscillates according to its characteristic natural frequencies, which are defined by its mass m, length L, and the spring constant k .

2.4 Analysis of Collective Coordinates (z_{CM} and z_{12})

The final stage of the experiment involves the decoupling of the system's equations of motion. By transforming the individual force data into collective coordinates, we can isolate the two fundamental normal modes that were previously superimposed in the individual sensor signals.

2.4.1 Coordinate Transformation Theory

Based on the linear relationship between the recorded forces and the displacements, the following coordinate transformations were applied:

- **Center of Mass Coordinate (z_{CM}):** Defined as the sum of the signals ($F_L + F_R$), isolating pure translational motion.
- **Relative/Rotational Coordinate (z_{12}):** Defined as the difference between the signals ($F_L - F_R$), isolating pure rotation about the center of mass.

2.4.2 Experimental Results

The Fast Fourier Transform (FFT) was applied to these transformed signals. The following table summarizes the frequencies identified for each motion type across the different trials:

Trial	ω_{sym} **(rad/s)**	**Error vs** ω_s	ω_{anti} **(rad/s)**	**Error vs** ω_a
1, RIGHT	12.7873	0.53%	19.0883	0.89%
1, LEFT	12.7173	1.08%	19.0760	0.82%
2, RIGHT	12.7758	0.62%	19.2684	1.84%
2, LEFT	12.7758	0.62%	19.0590	0.73%

Table 6: Calculated frequencies for decoupled coordinates. Reference values: $\omega_s = 12.856$ rad/s, $\omega_a = 18.92$ rad/s.

2.4.3 Discussion and Interpretation

The transformation to collective coordinates successfully decoupled the complex oscillations. As observed in the spectral analysis, the z_{CM} coordinate provided a single dominant peak corresponding to the symmetric frequency (ω_s), while the z_{12} coordinate isolated the antisymmetric frequency (ω_a). This proves that the normal modes are the natural basis for describing the motion of the coupled rigid body system. The effectiveness of this isolation is clearly displayed in figures hosted on drive [2], where the transformed temporal signals show the characteristic independent oscillations of each mode.

It is noteworthy that in trial "**1, LEFT**", where individual sensors previously failed to clearly identify the antisymmetric mode (errors over 32% in Section 3), the collective coordinate method managed to recover the correct frequency ($\omega_{anti} = 19.0760$ rad/s) with an error of only 0.82%. This demonstrates that combining the information from both sensors significantly increases the signal-to-noise ratio and eliminates the bias caused by the initial conditions of the release.

That way, the experimental evidence confirms that the motion of the bar is a superposition of two independent harmonic oscillators. By choosing the appropriate coordinates (z_{CM} and z_{12}), the equations of motion become uncoupled, allowing for a precise determination of the system's characteristic parameters regardless of the initial perturbation point.

3 Conclusion

The experimental study of coupled oscillations in a rigid bar system has successfully demonstrated the fundamental principles of classical mechanics regarding normal modes and coordinate transformation. Through the application of this three distinct methodologies (direct experimental measurement, sinusoidal fitting, and Fast Fourier Transform (FFT)) we achieved a high level of consistency in determining the characteristic frequencies of the system.

The results confirm that the motion of the coupled system is a linear superposition of two independent harmonic oscillators: the symmetric mode (ω_s) and the antisymmetric mode (ω_a). While a deviation was observed in the theoretical ratio ω_a/ω_s (obtaining ≈ 1.5 instead of the ideal $\sqrt{3} \approx 1.732$), this was conclusively attributed to the practical constraints of the experimental setup, specifically the internal positioning of the springs relative to the ends of the bar.

However, the most significant finding of this work lies in the analysis of collective coordinates (z_{CM} and z_{12}). By transforming the temporal force signals, we were able to mathematically decouple the equations of motion, effectively isolating each normal mode. This method proved to be remarkably robust, as evidenced by the successful recovery of the antisymmetric frequency

in Trial "**1, LEFT**", a case where individual sensor signals failed to distinguish the mode due to insufficient excitation. The reduction of relative errors to values below 1% in the decoupled analysis underscores the precision of using a normal basis for describing complex mechanical systems.

In summary, this laboratory session validates that spectral analysis and coordinate transformation are powerful tools for decomposing complex, coupled motions into simpler, independent components, providing a clear experimental proof of the superposition principle in rigid body dynamics.

4 Appendix

4.1 Graphics and fittings

Due to the large volume of generated data, this section presents a representative selection of the most significant plots. The complete repository, all individual sensor FFT spectra, sinusoidal fittings, and collective coordinate plots for every trial are also hosted and available for full review in the drive [2].

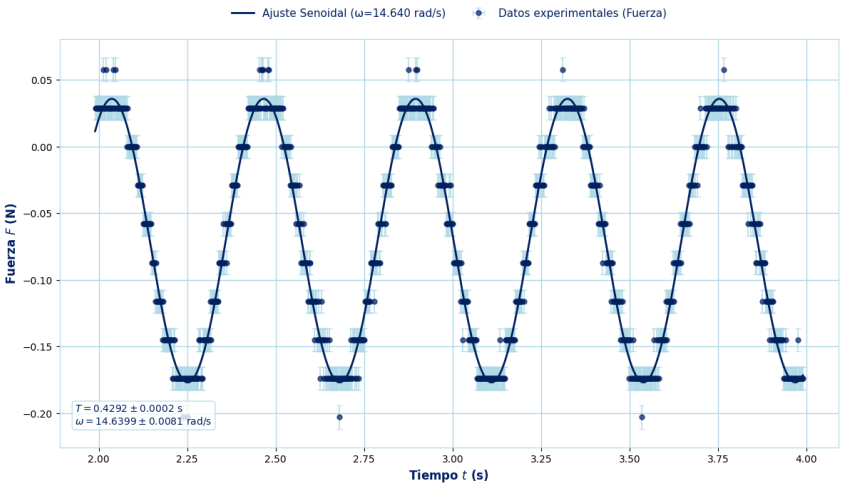


Figure 1: Experimental force measurements and sinusoidal regression for the first mode of oscillation.

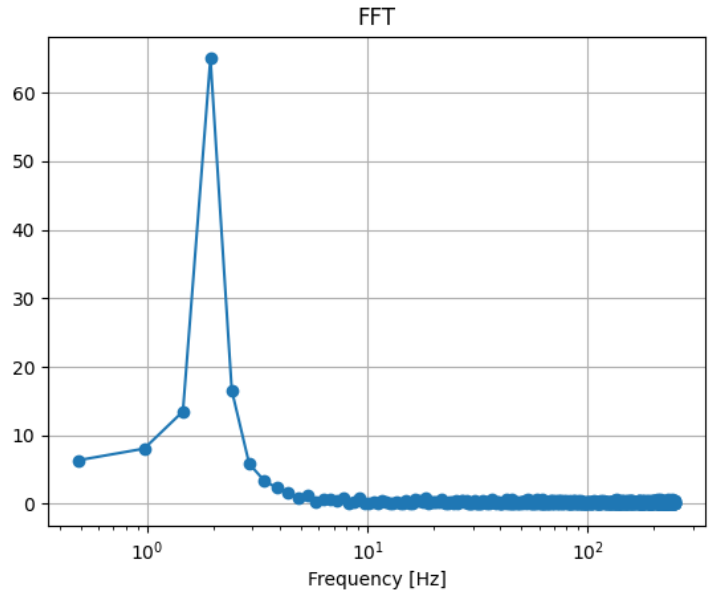


Figure 2: FFT spectral analysis of the force measured in the first oscillation mode. The dominant peak represents the system's characteristic frequency, while the rapidly decaying values at higher frequencies confirm the absence of significant higher-order harmonics or mechanical noise

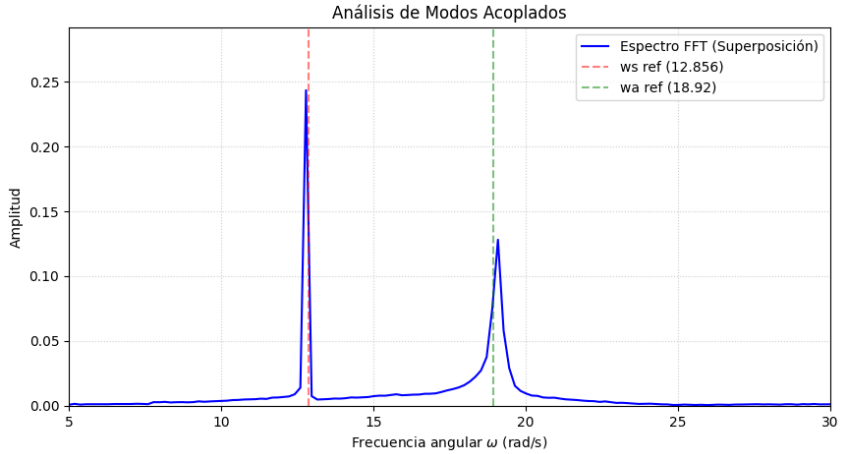


Figure 3: Frequency spectrum of the individual force sensor. Both normal modes are visible, with the primary peak corresponding to the symmetric mode (ω_s) and the secondary peak to the antisymmetric mode (ω_a)

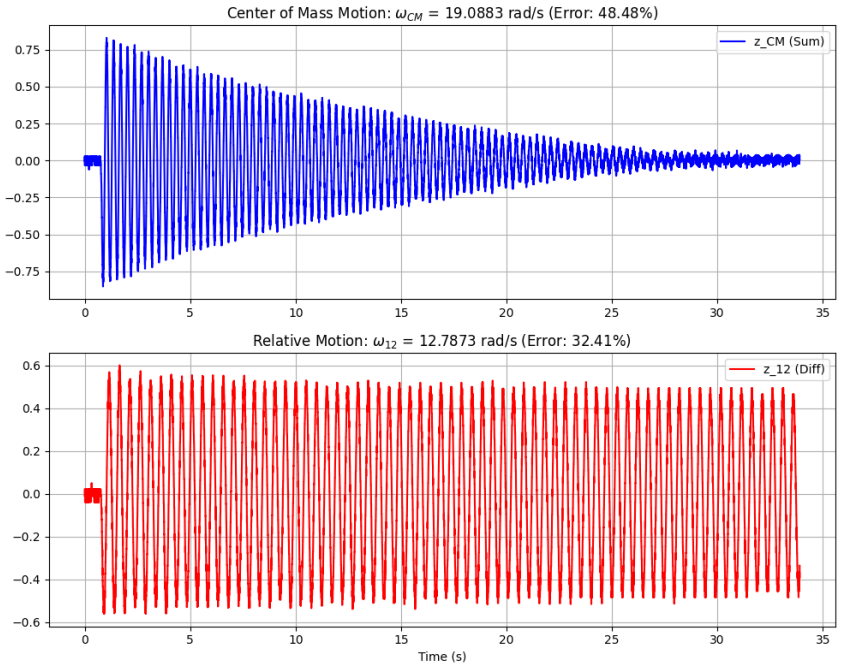


Figure 4: ime-domain representation of the decoupled collective coordinates. Although the labels are swapped due to sensor polarity, the signal z_{12} (red) clearly isolates the low-frequency symmetric mode, while z_{CM} (blue) isolates the high-frequency antisymmetric mode.

4.2 Data tables

4.2.1 Determination of the period and ferquency in pure harmonic oscillaions

5.2.1.1. Oscillation with a fixed end

	Experiment1	**Experiment 2**	**Experiment 3**
T_D (s)	$0,426 \pm 0,011$	$0,431 \pm 0,033$	$0,436 \pm 0,011$
T_I (s)	$0,4317 \pm 0,0021$	$0,433 \pm 0,012$	$0,4303 \pm 0,0019$
$\overline{T_o}$ (s)	$0,4290 \pm 0,0055$	$0,432 \pm 0,018$	$0,4332 \pm 0,0058$
ω_o (rad/s)	$14,65 \pm 0,19$	$14,55 \pm 0,59$	$14,50 \pm 0,19$
f_o (Hz)	$2,331 \pm 0,030$	$2,315 \pm 0,095$	$2,308 \pm 0,031$

Table 7: Periods of the left and right springs and their average, $\overline{T}$, with their corresponding errors, obtained experimentally employing the first oscillation mode. The angular frequency ω and f were also calculated for each experience.

	Experiment1	**Experiment 2**	**Experiment 3**
T_o (s)	$0,4294 \pm 0,0002$	$0,4292 \pm 0,0002$	$0,4303 \pm 0,0001$
ω_o (rad/s)	$14,6333 \pm 0,0061$	$14,6399 \pm 0,0081$	$14,6003 \pm 0,0041$
f_o (Hz)	$2,3288 \pm 0,0011$	$2,3299 \pm 0,0011$	$2,32396 \pm 0,00054$

Table 8: Period T, angular frequency ω and f with their corresponding errors, calculated by the sinusoidal fitting.

	Experiment 1	**Experiment 2**	**Experiment 3**
T_o (s)	$0,42400 \pm 0,00010$	$0,41700 \pm 0,00010$	$0,43018 \pm 0,00010$
ω_o (rad/s)	$14,8188 \pm 0,0035$	$15,0676 \pm 0,0035$	$14,6059 \pm 0,0035$
f_o (Hz)	$2,35849 \pm 0,00056$	$2,39808 \pm 0,00056$	$2,3246 \pm 0,00056$

Table 9: Period $\overline{T_o}$, angular frequency ω_o and f_o with their corresponding errors, using the FFT method.

5.2.1.2. Symmetrical oscillation

	Experiment1	**Experiment 2**	**Experiment 3**
T_D (s)	$0,4915 \pm 0,0028$	$0,4921 \pm 0,0022$	$0,4832 \pm 0,0035$
T_I (s)	$0,4861 \pm 0,0032$	$0,4919 \pm 0,0023$	$0,4877 \pm 0,0028$
$\overline{T_s}$ (s)	$0,4888 \pm 0,0021$	$0,4920 \pm 0,0016$	$0,4855 \pm 0,0023$
ω_s (rad/s)	$12,854 \pm 0,056$	$12,771 \pm 0,041$	$12,943 \pm 0,060$
f_s (Hz)	$2,0458 \pm 0,0089$	$2,0326 \pm 0,0065$	$2,060 \pm 0,010$

Table 10: Periods of the left and right springs and their average, $\overline{T}$, with their corresponding errors, obtained experimentally employing the symmetrical oscillation mode. The angular frequency ω and f were also calculated for each experience.

	Experiment1	**Experiment 2**	**Experiment 3**
T_D (s)	$0,4926 \pm 0,0002$	$0,4909 \pm 0,0002$	$0,4920 \pm 0,0002$
T_I (s)	$0,4899 \pm 0,0002$	$0,4919 \pm 0,0002$	$0,4916 \pm 0,0001$
$\overline{T_s}$ (s)	$0,49125 \pm 0,00014$	$0,49140 \pm 0,00014$	$0,49180 \pm 0,00011$
ω_s (rad/s)	$12,7902 \pm 0,0037$	$12,7863 \pm 0,0037$	$12,7759 \pm 0,0029$
f_s (Hz)	$2,03562 \pm 0,00059$	$2,03500 \pm 0,00059$	$2,03335 \pm 0,00046$

Table 11: Period T, angular frequency ω and f with their corresponding errors, calculated by the sinusoidal fitting.

	Experiment 1	**Experiment 2**	**Experiment 3**
$\overline{T_s}$ (s)	$0,49717 \pm 0,00010$	$0,49349 \pm 0,00010$	$0,49159 \pm 0,00009$
ω_s (rad/s)	$12,6378 \pm 0,0025$	$12,7322 \pm 0,0025$	$12,7812 \pm 0,0025$
f_s (Hz)	$2,0114 \pm 0,0004$	$2,0264 \pm 0,0004$	$2,0342 \pm 0,0004$

Table 12: Average period $\overline{T_s}$, angular frequency ω_s and f_s with their corresponding errors, using the FFT method.

5.2.1.3. Antisymmetric oscillation

	Experiment1	**Experiment 2**	**Experiment 3**
T_D (s)	$0,3349 \pm 0,0057$	$0,3296 \pm 0,0030$	$0,3296 \pm 0,0020$
T_I (s)	$0,345 \pm 0,012$	$0,3252 \pm 0,0036$	$0,3292 \pm 0,0023$
$\overline{T_a}$ (s)	$0,3398 \pm 0,0068$	$0,3274 \pm 0,0023$	$0,3294 \pm 0,0015$
ω_a (rad/s)	$18,49 \pm 0,37$	$19,19 \pm 0,14$	$19,075 \pm 0,088$
f_a (Hz)	$2,943 \pm 0,059$	$3,054 \pm 0,022$	$3,036 \pm 0,014$

Table 13: Periods of the left and right springs and their average, $\overline{T}$, with their corresponding errors, obtained experimentally employing the antisymmetric oscillation mode. The angular frequency ω and f were also calculated for each experience.

	Experiment1	**Experiment 2**	**Experiment 3**
T_D (s)	$0,3289 \pm 0,0005$	$0,3290 \pm 0,0007$	$0,3294 \pm 0,0003$
T_I (s)	$0,3282 \pm 0,0004$	$0,3290 \pm 0,0008$	$0,3297 \pm 0,0003$
$\overline{T_a}$ (s)	$0,32855 \pm 0,00032$	$0,32900 \pm 0,00053$	$0,32955 \pm 0,00021$
ω_a (rad/s)	$19,124 \pm 0,019$	$19,098 \pm 0,031$	$19,066 \pm 0,012$
f_a (Hz)	$3,0437 \pm 0,0030$	$3,0395 \pm 0,0049$	$3,0344 \pm 0,0020$

Table 14: Period T, angular frequency ω and f with their corresponding errors, calculated by the sinusoidal fitting.

	Experiment 1	**Experiment 2**	**Experiment 3**
$\overline{T_a}$ (s)	$0,32991 \pm 0,000040$	$0,32647 \pm 0,000040$	$0,32142 \pm 0,000040$
ω_a (rad/s)	$19,0454 \pm 0,0025$	$19,2458 \pm 0,0025$	$19,5482 \pm 0,0025$
f_a (Hz)	$3,0312 \pm 0,00040$	$3,0631 \pm 0,00040$	$3,1112 \pm 0,00040$

Table 15: Period $\overline{T_a}$, angular frequency ω_a and f_a with their corresponding errors, using the FFT method.

4.3 Uncertainties

4.3.1 General Uncertainties Calculations

As established in measurement uncertainty analysis, Type A uncertainty, denoted as $u_A(x)$, quantifies the statistical variation associated with repeated observations and is calculated using the equation:

$$u_A(x) = \frac{\sigma(x)}{\sqrt{N}}, \tag{2}$$

where $\sigma(x)$ represents the standard deviation and N the number of observations. On the other hand, Type B uncertainty, $u_B(x)$, is derived from the estimated accuracy of the measuring instrument and is obtained from:

$$u_B(x) = \frac{\Delta_x}{\sqrt{12}}, \tag{3}$$

denotes the instrument's resolution ($\delta_V = 0.01$ V for voltage measurements and $\delta_T = 0.001$ s for the photogate period measurements). The combined standard uncertainty, $u_C(x)$, is subsequently expressed as the root sum of squares of these components:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)}. \tag{4}$$

For derived quantities obtained through functional relationships $y = f(x_1, x_2, \ldots, x_n)$, the uncertainty was propagated using:

$$u_C(y) = \sqrt{\sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}\right)^2 u_C(x_i)^2}. \tag{5}$$

Finally, the expanded uncertainty is defined as:

$$U_C(x) = k \cdot u_C(x), \tag{6}$$

where k represents the coverage factor. Although the GUM [3] allows for various approaches depending on the sample size, we have adopted a coverage factor of $k = 2$. This choice is justified by the large volume of data collected (approximately 15000 points per experiment). With such a significant sample size, the distribution of the results effectively converges to a normal distribution. In this context, a factor of $k = 2$ provides a confidence level of approximately 95% [4].

4.3.2 Direct measurements

The data was collected in the laboratory by two PASCO sensors, which presented a resolution of $\Delta F = 0,03\ N$ and $\Delta t = 0,002\ s$. Consequently,

$$u_C(t) = \frac{\Delta_t}{\sqrt{12}} = 0,00058\ \text{ s}, \tag{7}$$

$$u_C(F) = \frac{\Delta_F}{\sqrt{12}} = 0,0087\ \text{ N}. \tag{8}$$

4.3.3 Specific Uncertainty Calculations

Uncertainty in ω and f from the experimental method and sinusoidal regression. For the magnitudes of ω and f derived from the period, i.e., using the experimental and sinusoidal method, their associated errors were obtained as follows.

$$u_C(\omega) = \frac{2\pi \cdot u_C(T)}{T^2}\ \text{ rad/s}, \tag{9}$$

$$u_C(f) = \frac{u_C(T)}{T^2}\ \text{ Hz}. \tag{10}$$

Uncertainty associated to a general average value $\overline{x}$. For every average value calculated from a dataset of N values, its error is given by

$$u_C(\overline{x}) = \frac{1}{N}\sqrt{u_c(x_1)^2 + u_C(x_2)^2 + ... + u_C(x_n)^2}. \tag{11}$$

Uncertainty in ω and T from FFT. The uncertainty associated with the angular frequencies obtained via Fast Fourier Transform (FFT) is governed by the spectral resolution of the transformation. Unlike direct temporal measurements, the precision of the FFT depends on the total acquisition time (T_{total}). The frequency resolution in Hertz is defined as $\Delta f = 1/T_{total}$. Consequently, the resolution in terms of angular frequency ($\Delta\omega$) is:

$$\Delta\omega = \frac{2\pi}{T_{total}} \tag{12}$$

In accordance with the criteria for digital instruments and rectangular distributions, the standard uncertainty $u_C(\omega)$ is derived from the spectral resolution as:

$$u_C(\omega) = \frac{\Delta\omega}{\sqrt{12}} \tag{13}$$

For the trials presented in 5, the spectral resolutions ($\Delta\omega$) range between 0.1382 rad/s and 0.2095 rad/s. Taking the worst-case scenario ($\Delta\omega = 0.2095$ rad/s), the standard uncertainty is:

$$u_C(\omega) \approx \frac{0.2095}{\sqrt{12}} \approx 0.0605\ \text{rad/s} \tag{14}$$

For the period, the associated error was calculated as follows.

$$u_C(T) = \frac{u_C(f)}{f^2}\ \text{ s}. \tag{15}$$

Uncertainty in Collective Coordinates (z_{CM} and z_{12}). In **Section 3.4**, the signals are constructed using linear combinations: $z_{CM} \propto F_L + F_R$ and $z_{12} \propto F_L - F_R$. Since the FFT is a linear operator, the frequency identified in the transformed signal inherits the same spectral resolution as the original force signals.

The uncertainty in the identified frequencies ω_{sym} and ω_{anti} is determined by the same spectral binning constraint:

$$u_C(\omega_{sym}) = u_C(\omega_{anti}) = \frac{\Delta\omega}{\sqrt{12}} \tag{16}$$

This confirms that the decoupling process does not add additional frequency uncertainty, provided that the sampling rate and total time remain constant for both sensors.

Relative Error Calculation. To evaluate the consistency of the results against the reference values ($\omega_s = 12.856$ rad/s and $\omega_a = 18.92$ rad/s), the relative error ε_r was calculated as:

$$\varepsilon_r = \frac{|\omega_{exp} - \omega_{ref}|}{\omega_{ref}} \cdot 100 \tag{17}$$

The results summarized in 6 show that for decoupled coordinates, these errors remain mostly below 1%, which is within the confidence interval defined by the spectral resolution.

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